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Preserving Proton Resolution Under Cryogenic Dynamic Nuclear Polarization for Biomolecular Solid-State NMR

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ABSTRACT

The combination of proton detection with cryogenic dynamic nuclear polarization (DNP) promises to unite the intrinsic sensitivity and resolution of ^{1}H-detected magic-angle spinning (MAS) NMR with the large signal enhancements afforded by hyperpolarization, but has long been considered impractical because of cryogenic line broadening and the limited spinning frequencies available under DNP conditions. Here, we demonstrate that high-resolution proton-detected spectra of proteins can indeed be obtained under cryogenic DNP conditions. Using extensive deuteration, selective $^{13}CHD_2$ methyl labeling, fast MAS and high magnetic fields, we show that site-resolved proton detection is preserved in both microcrystalline proteins and amyloid fibrils. Quantitative linewidth analysis reveals that, despite broader resonances, proton coherence lifetimes remain remarkably long because high magnetic fields, fast MAS, and extensive proton dilution efficiently suppress the homogeneous contribution, leaving the observed linewidth dominated by refocusable inhomogeneous broadening. The resulting sensitivity enables structurally informative proton-detected ^{1}H-^{1}H correlation spectroscopy under DNP conditions. These results establish proton-detected DNP as a practical methodology for biomolecular MAS NMR and extend structural studies to challenging protein assemblies limited by sensitivity, spectral crowding, or sample availability.

For more than two decades, solid-state NMR spectroscopy with magic-angle spinning (MAS) has provided unparalleled insights into the structure and dynamics of insoluble and non-crystalline biomolecular systems [1]. Despite these successes, biomolecular MAS NMR remains fundamentally limited by sensitivity, as detection typically relies on low-gyromagnetic ratio (γ) nuclei such as ^{13}C and ^{15}N. As a result, the development of strategies to boost sensitivity has been a central theme in advancing the field.

One strategy is the detection of high-γ nuclei such as ^{1}H [2]. Traditionally, this was possible only at relatively slow MAS rates (10–20 kHz) and required extensive proton dilution, which partly offset the inherent sensitivity gains [3]. The advent of

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probes capable of faster spinning rates (exceeding 100 kHz today) has transformed this situation [4]. Fast MAS efficiently reduces the homonuclear ^{1}H-^{1}H dipolar couplings that dominate the solid state, narrowing ^{1}H resonance lines and enabling the practical use of ^{1}H detection in biomolecular NMR. For a given sample volume, this provides an intrinsic detection-sensitivity gain of up to $(\gamma_H/\gamma_C)^{3/2}\sim8$ and $(\gamma_H/\gamma_N)^{3/2}\sim31$ relative to detection of ^{13}C and ^{15}N, respectively. This has reduced sample requirements and improved practical sensitivity for large protein assemblies, fibrils and membrane systems, complementing the extensive body of work based on conventional ^{13}C and ^{15}N detection.

A second strategy for enhancing sensitivity is dynamic nuclear polarization (DNP), which provides one-to-two order of magnitude signal enhancements by transferring electron spin polarization to nuclei at cryogenic temperatures under microwave irradiation [5]. DNP has opened entirely new regimes for solid-state NMR of materials and pharmaceutical systems [6] and is increasingly applied in structural biology [7, 8], enabling atomic-level studies of amyloids, membrane proteins, and even cellular preparations that were previously inaccessible by conventional methods [9–18].

In principle, combining ^{1}H detection with cryogenic DNP should offer the best of both worlds: the extraordinary sensitivity gains of hyperpolarization and the high resolution and intrinsic sensitivity advantages of proton-detected MAS NMR. Yet, until now, this combination has been considered impractical. Cryogenic conditions broaden resonances due to conformational heterogeneity [19–21] and paramagnetic effects, while MAS rates available in DNP probes were too slow to average homonuclear couplings. As a result, resolved ^{1}H-detected spectra of proteins under DNP conditions had not been demonstrated, and the feasibility of such experiments remained an open question.

Only very recently, the advent of commercial DNP probes capable of spinning at 40–60 kHz has enabled pioneering proof-of-principle studies, showing that ^{1}H detection can enhance 2D correlations in inorganic [22] and organic powders [23, 24]. Importantly, MAS probes operated under cryogenic DNP conditions reach lower maximum spinning rates than their room-temperature counterparts (~40 vs. 60 kHz for 1.3 mm rotors, ~60 vs. 100 kHz for 0.7 mm rotors), owing to the unfavorable effect of higher nitrogen density at low temperature. However, MAS at 40 kHz is already sufficient to efficiently average ^{1}H-^{1}H homonuclear dipolar couplings when combined with extensive deuteration, thereby providing the resolution required for biomolecular studies [25].

More recently, high-resolution proton-detected spectra of selectively protonated aromatic residues were shown to be preserved at cryogenic temperatures under fast MAS in the absence of DNP [26]. These results suggested that the combination of sparse protonation and efficient dipolar averaging can preserve high-resolution proton detection despite cryogenic temperatures. Whether this strategy could be extended to cryogenic DNP conditions, where the presence of paramagnetic polarizing agents and the lower MAS frequencies available in DNP probes introduce additional sources of broadening, remained an open question.

Here, we demonstrate that this strategy can indeed be extended to proton-detected cryogenic DNP. Using extensive deuteration, selective $^{13}CHD_2$ labeling, fast MAS at 40 kHz, high magnetic fields, and efficient hyperpolarization, we show that high-resolution proton-detected spectra can be preserved under cryogenic DNP conditions. By examining two benchmark protein systems, we establish that the sensitivity of DNP can be constructively combined with the intrinsic resolution and sensitivity of ^{1}H detection, defining a new experimental regime for biomolecular MAS NMR.

To assess the performance of the approach in different structural contexts, we selected two benchmark systems representing distinct classes of biomolecular solids. The SH3 domain of α-spectrin (Figure 1A) forms well-characterized microcrystals that have become a standard model system for biomolecular MAS NMR [27], whereas the 8.6 kDa prion-forming domain (residues 218–289) of the functional amyloid HET-s (Figure 1B) adopts a well-characterized β-solenoid structural fold [28] with a rigid hydrophobic core [29] that minimizes line broadening at cryogenic temperatures [30]. These systems differ markedly in molecular architecture, packing and dynamics, providing a stringent test of whether proton detection remains practical under cryogenic DNP conditions.

To maximize resolution under these conditions, we employed a metabolic precursor during protein production in a deuterated medium [31] to restrict the observable protons to $^{13}CHD_2$ in pro-S-methyl groups of isoleucine, leucine and valine (ILV) residues. This labeling scheme dramatically reduces the proton density while preserving structurally informative methyl probes. Following expression in a fully deuterated medium, complete back-exchange and selective ILV labeling, the residual proton contents are approximately 29% for HET-s and 26% for SH3.

Samples were prepared by soaking both protein samples with M-TinyPol [32], a water-soluble polarizing agent, and glycerol (Figure 1). To benchmark the effect of cryogenic DNP, we compared each protein under optimized room-temperature and cryogenic conditions. Room-temperature reference spectra were acquired at ~298 K using a conventional 1.3 mm MAS probe operated at its maximum spinning frequency (60 kHz) together with a J-based (H)CH coherence-transfer experiment, which provides the highest sensitivity under ambient conditions. Cryogenic DNP experiments were performed at 100 K and 40 kHz MAS using the corresponding dipolar-based (H)CH experiment appropriate under these conditions. The comparison therefore benchmarks the optimal implementation of conventional room-temperature proton-detected MAS NMR against that obtained under cryogenic DNP conditions, rather than assessing the influence of any individual experimental parameter.

(H)CH spectra recorded at room temperature yielded well-resolved signals for both proteins. For SH3, all expected ILV methyl resonances were detected except for L31 and V46, consistent with previous room-temperature spectra of this system [33]. For HET-s(218–289), methyl signals originating from the rigid amyloid core were all observed. Upon cooling to 100 K under DNP conditions, the resonances broaden to different extents while remaining sufficiently resolved for site-specific analysis (Figure 2B, F). The extent of broadening varies from

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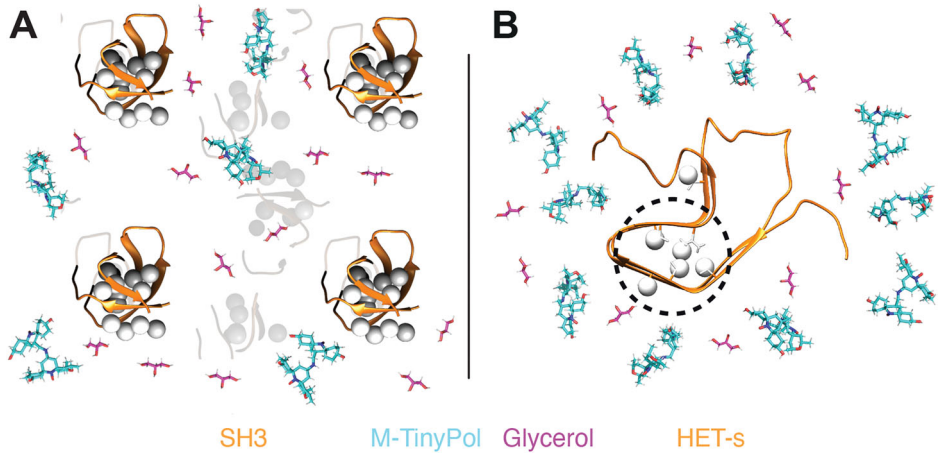


FIGURE 1 | Benchmark protein systems and sample architecture for proton-detected DNP MAS NMR. (A) Schematic representation of the SH3 microcrystalline DNP sample, including a representative SH3 molecule (orange, derived from PDB entry 1U06) surrounded by M-TinyPol biradicals (light blue) and glycerol molecules (magenta). ILV-$^{13}CHD_2$-labeled methyl groups are shown as white spheres. (B) Sketch of the HET-s(218–289) fibrillar sample, including a top view of one HET-s(218–289) monomer within the fibrillar assembly (orange, derived from PDB entry 2KJ3) surrounded by M-TinyPol biradicals (light blue) and glycerol molecules (magenta). ILV-$^{13}CHD_2$-labelled methyl groups are shown as white spheres.

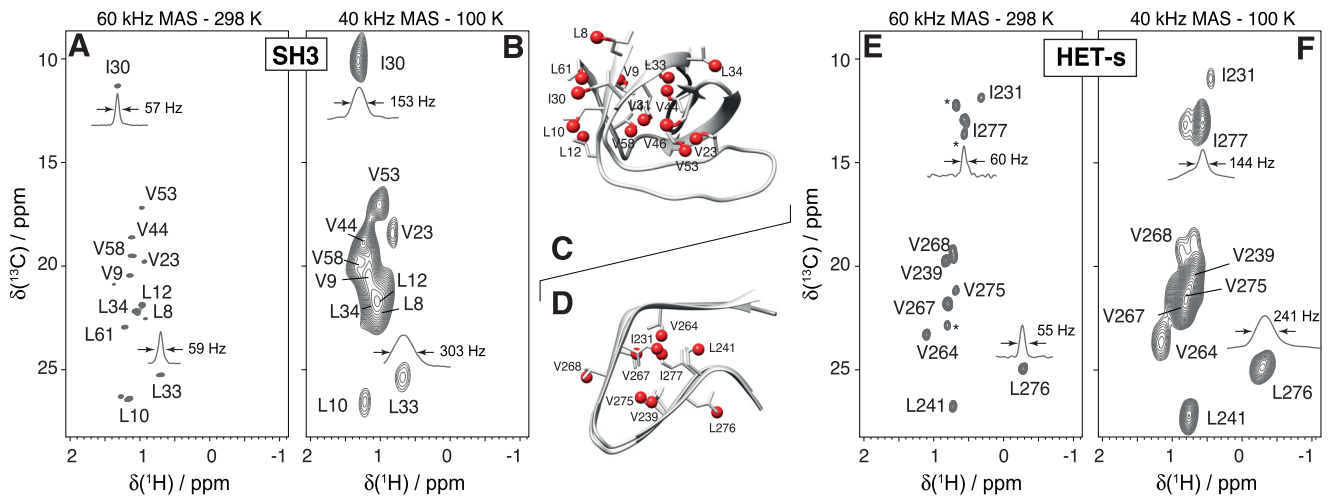


FIGURE 2 | Evolution of proton-detected methyl spectra from room temperature to cryogenic DNP conditions. (A and B) Proton-detected (H)CH spectra of selectively ILV-$^{13}CHD_2$-labeled microcrystalline SH3 domain recorded under conventional room-temperature MAS (60 kHz) (A) and under cryogenic DNP conditions (40 kHz MAS, 100 K) (B). (C) Structure of the SH3 domain (PDB entry 1U06) and (D) structure of the rigid core of one HET-s(218–289) monomer within the fibrillar assembly (PDB entry 2KJ3), with selectively ILV-$^{13}CHD_2$-labelled methyl groups shown as red spheres. (E and F) Proton-detected (H)CH spectra of selectively ILV-$^{13}CHD_2$-labeled HET-s(218–289) amyloid fibrils recorded under conventional room-temperature MAS (60 kHz) (E) and under cryogenic DNP conditions (40 kHz MAS, 100 K) (F). Unassigned signals from the flexible and unstructured N- and C-termini of HET-s(218-289) are marked by asterisks.

site to site, with methyl linewidths ranging from approximately 150–300 Hz in SH3 and from 150–250 Hz in HET-s(218–289), reflecting differences in the local structural environment and conformational heterogeneity in the frozen state. The magnitude of the broadening varies from residue to residue but does not exhibit any simple structural trend (Table SI1). Nevertheless, the overall loss of spectral resolution remains surprisingly modest. In SH3, all room-temperature resonances remain identifiable under cryogenic DNP conditions and, with one exception, retain essentially unchanged chemical shifts. V23, located in the long surface loop, is weaker and slightly shifted, consistent with a greater sensitivity of this flexible surface region to cooling. Similarly, in HET-s, all nine expected methyl correlations from the rigid fibril core are retained under cryogenic DNP conditions, whereas resonances from the flexible N- and C-terminal regions are absent. These results demonstrate that, despite the inevitable linewidth increase accompanying cryogenic DNP, simple 2D proton-detected (H)CH experiments preserve sufficient resolution for site-specific structural analysis of both crystalline proteins and amyloid fibrils.

The preservation of proton resolution under cryogenic DNP conditions relies on two experimentally controllable parameters: sufficiently fast MAS to average the residual homonuclear dipolar

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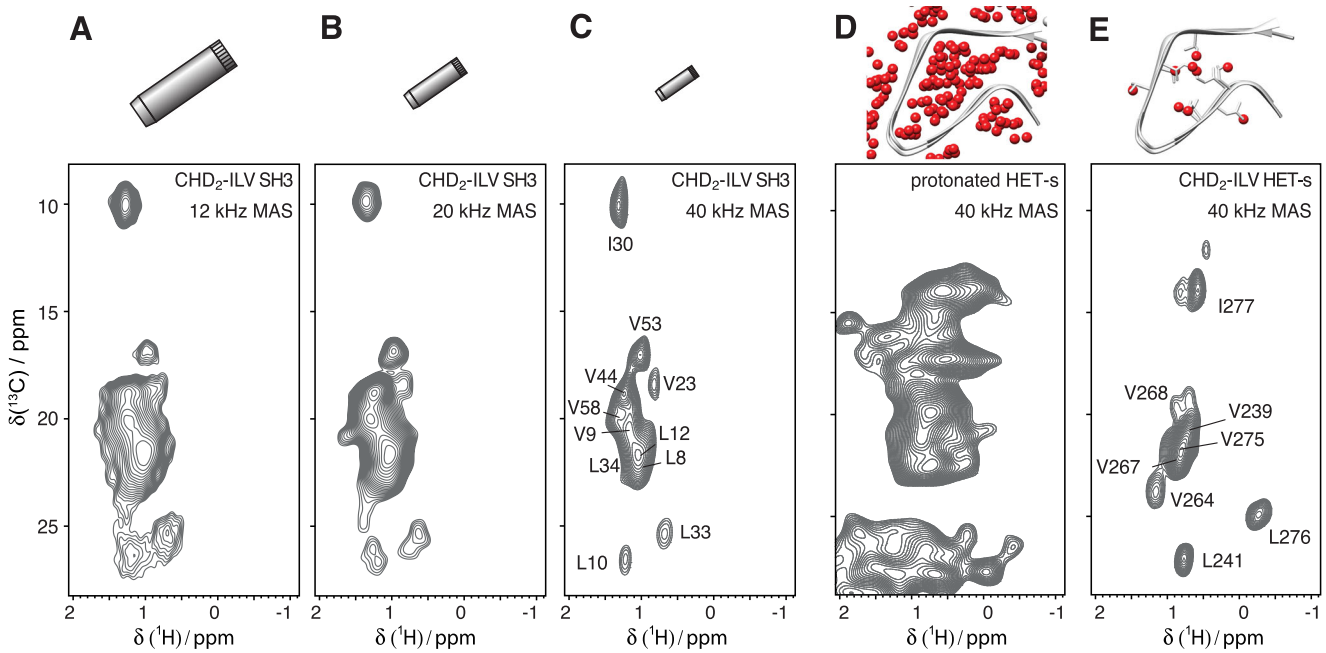


FIGURE 3 | Experimental conditions required to preserve proton resolution under cryogenic DNP conditions. (A–C) Proton-detected (H)CH spectra of selectively ILV-$^{13}CHD_2$-labeled SH3 microcrystals recorded under cryogenic DNP conditions at 12 kHz (A), 20 kHz (B), and 40 kHz MAS (C). Although all spectra were acquired using the same 1.3 mm DNP probe, the rotor cartoons indicate the MAS frequencies typically associated with commercial 3.2, 1.9, and 1.3 mm DNP probes. Increasing the MAS frequency progressively suppresses residual ^{1}H-^{1}H dipolar couplings, resulting in substantial line narrowing. (D and E) Proton-detected (H)CH spectra of HET-s(218–289) acquired under otherwise identical cryogenic DNP conditions at 40 kHz MAS using fully protonated, uniformly ^{13}C-labelled protein (D) and selectively ILV-$^{13}CHD_2$-labeled protein (E). The accompanying schematics illustrate the corresponding proton densities. Extensive proton dilution is essential for preserving site-resolved proton detection at the MAS frequencies currently accessible under cryogenic DNP conditions.

couplings, and extensive proton dilution to reduce their magnitude. These two effects are illustrated independently in Figure 3. First, the influence of MAS frequency was examined by recording proton-detected (H)CH spectra of selectively ILV-$^{13}CHD_2$-labeled SH3 microcrystals at 12, 20, and 40 kHz MAS under otherwise identical cryogenic DNP conditions (Figure 3A–C). Although all spectra were acquired using the same 1.3 mm DNP probe, the selected spinning frequencies correspond to those typically associated with commercial 3.2, 1.9, and 1.3 mm DNP probes, respectively. Increasing the MAS frequency progressively narrows the proton resonances, reflecting increasingly efficient averaging of residual ^{1}H-^{1}H dipolar couplings. Second, the role of proton dilution was evaluated under otherwise identical cryogenic DNP conditions by comparing selectively ILV-$^{13}CHD_2$-labelled and fully protonated, uniformly ^{13}C-labeled HET-s(218–289) samples (Figure 3D, E). Whereas the spin-diluted sample yields narrow, site-resolved methyl correlations, the fully protonated sample is devoid of site-specific resolution. Thus, although fast MAS efficiently suppresses the residual homonuclear dipolar couplings, extensive proton dilution remains essential for preserving high-resolution proton detection at the spinning frequencies currently accessible under cryogenic DNP conditions.

As expected from the favorable detection sensitivity of protons, direct comparison with a conventional ^{13}C-detected HETCOR experiment confirms an additional sensitivity advantage for the ^{1}H-detected scheme under DNP conditions (Figure SI2) [34].

Because the observed proton linewidth contains both homogeneous and inhomogeneous contributions, its physical origin cannot be inferred from the linewidth alone. To further dissect the physical origin of the residual proton linewidth under cryogenic DNP conditions, we compared the observed linewidths with the corresponding homogeneous and inhomogeneous contributions derived from bulk transverse relaxation measurements (T_2') (Figure 4). Room-temperature measurements were performed on both undoped and DNP-formulated SH3 samples at 40 and 60 kHz MAS, whereas cryogenic measurements were acquired on the DNP sample at 10 and 40 kHz MAS. This experimental design separates the respective effects of the polarizing agent, conformational freezing and MAS frequency on proton linewidths. Analogous room-temperature and cryogenic T_2' measurements on HET-s(218–289) support the same overall picture (Figure SI3).

At room temperature, addition of the polarizing agent produces a moderate increase in the homogeneous linewidth, indicating that paramagnetic broadening makes a relatively minor contribution. By contrast, cooling to 100 K substantially increases the observed linewidth, predominantly through an increase in the inhomogeneous (refocusable) contribution associated with conformational freezing. Increasing the MAS frequency from 12 to 40 kHz progressively suppresses the residual homogeneous broadening arising from ^{1}H-^{1}H dipolar couplings, leading to a marked increase in proton coherence lifetimes. As a result, the dominant origin of the linewidth changes with MAS frequency: at 10 kHz the observed linewidth is governed primarily by

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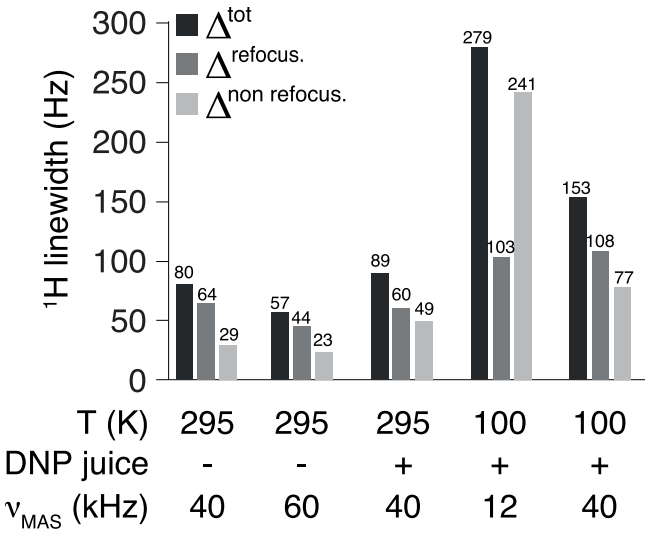


FIGURE 4 | Physical origin of proton linewidths under cryogenic DNP conditions. Comparison of the observed proton linewidths (Δ^{tot}, dark bars), non-refocusable linewidths derived from bulk transverse relaxation measurements ($\Delta^{non\ refocus} = 1/\pi T_2'$, light bars), and the corresponding inhomogeneous contributions ($\Delta^{refocus}$, grey bars) for residue I30 in the selectively ILV-$^{13}CHD_2$-labeled SH3 domain under different experimental conditions. Room-temperature measurements were performed at 40 and 60 kHz MAS in the absence (DNP juice −) and presence (DNP juice +) of M-TinyPol, whereas cryogenic measurements were acquired at 100 K under DNP conditions at 12 and 40 kHz MAS. The data show that increasing the MAS frequency changes the dominant broadening mechanism from homogeneous dipolar broadening at slow MAS to predominantly refocusable inhomogeneous broadening at fast MAS. The decomposition of the observed linewidth into homogeneous and inhomogeneous contributions is described in the Supporting Information.

homogeneous broadening, whereas at 40 kHz the remaining linewidth is dominated by inhomogeneous broadening.

Overall, the average methyl linewidth increases from approximately 60 Hz at room temperature to ~250 Hz under cryogenic DNP conditions. Remarkably, under the combined conditions of high magnetic field, fast MAS and extensive deuteration, the homogeneous contribution remains comparatively small. It amounts to only ~75 Hz, corresponding to a proton T_2' exceeding 6 ms at 100 K, and is therefore only modestly larger than the ~40 Hz measured at room temperature despite the presence of the paramagnetic polarizing agent. Consequently, once the homogeneous broadening arising from residual dipolar couplings has been efficiently suppressed, the additional linewidth observed under cryogenic DNP conditions is dominated by refocusable inhomogeneous broadening associated with conformational freezing rather than by irreversible loss of proton coherence. Thus, although cryogenic DNP inevitably broadens the observed resonances, it preserves unexpectedly long proton coherence lifetimes, allowing the large sensitivity gains afforded by DNP to be exploited without sacrificing high-resolution proton detection.

Having established that high-resolution proton detection is preserved under cryogenic DNP conditions, we next evaluated the corresponding sensitivity gains. Although the microwave-induced enhancement factors exceeded ~70 for both samples, the enhancement factor alone does not reflect the overall experimental gain because absolute sensitivity also depends on cryogenic relaxation effects (i.e., longer T_1 relaxation times, shorter signal lifetimes T_2*, increased Boltzmann polarization, altered molecular dynamics) or the influence of radicals (paramagnetic quenching [35, 36] and depolarization [37, 38]). Direct comparison of the corresponding room-temperature and DNP experiments reveals an overall sensitivity gain of up to ~6-fold under cryogenic DNP conditions (see Figure SI4).

The enhanced sensitivity afforded by DNP renders intrinsically insensitive multidimensional proton-detected experiments accessible, whereas they would otherwise require prohibitively long acquisition times. In the present work, we exploited this enhanced sensitivity to record structurally informative proton–proton correlation spectra, which benefit simultaneously from the increased sensitivity and from the stronger dipolar couplings present at cryogenic temperatures. Cryogenic temperatures suppress molecular motions that partially average out dipolar couplings, leading to stronger effective ^{1}H-^{1}H interactions. Molecular-dynamics and NMR relaxation analyses of HET-s side chains show that methyl and χ_1/χ_2 reorientations occur on subnanosecond to microsecond time scales at 300 K, leading to order parameters S^2 of ≈ 0.2-0.3 for most core residues [39]. Upon cooling to 100 K, these motions are largely frozen, and the corresponding order parameters typically rise to ≈ 0.6–0.8 [40]. This behavior is directly reflected in our spectra, where dipolar-based ^{1}H-^{13}C transfers, which are only 20%–30% as efficient as scalar transfers at room temperature, become dominant under DNP conditions. Together with the enhanced sensitivity afforded by DNP, these stronger dipolar interactions enable efficient proton-detected ^{1}H-^{1}H correlation spectroscopy. A 2D [^{1}H,^{1}H]-RFDR [41] spectrum acquired in only 8 h shows strong signals with sufficient resolution to identify all ^{13}C-labeled methyl groups and a large number of through-space correlations within the fibril core (Figure 5A, B). From these data, we derived distance restraints up to 6 Å that agree with the known 3D structural model.

Our observations complement recent DNP-enhanced studies demonstrating that cryogenic linewidths can provide detailed information on frozen conformational ensembles and local structural heterogeneity through selectively labeled residues, particularly isoleucine side chains [42]. Whereas those studies intentionally exploited conformational broadening as a structural probe, our work shows that extensive deuteration, sparse $^{13}CHD_2$ methyl protonation, fast MAS, and high magnetic fields preserve high-resolution proton detection without eliminating the structural information encoded in cryogenic linewidths. Quantitative linewidth analysis further reveals why this is possible: the combination of high magnetic field, fast MAS, and extensive proton dilution suppresses the homogeneous contribution to the linewidth sufficiently that the remaining broadening is dominated by refocusable inhomogeneous contributions, leaving unexpectedly long proton coherence lifetimes despite the presence of paramagnetic polarizing agents. These results establish proton-detected DNP as a practical methodology for biomolecular solid-state NMR and identify fast MAS together with extensive proton dilution as the key experimental ingredients required

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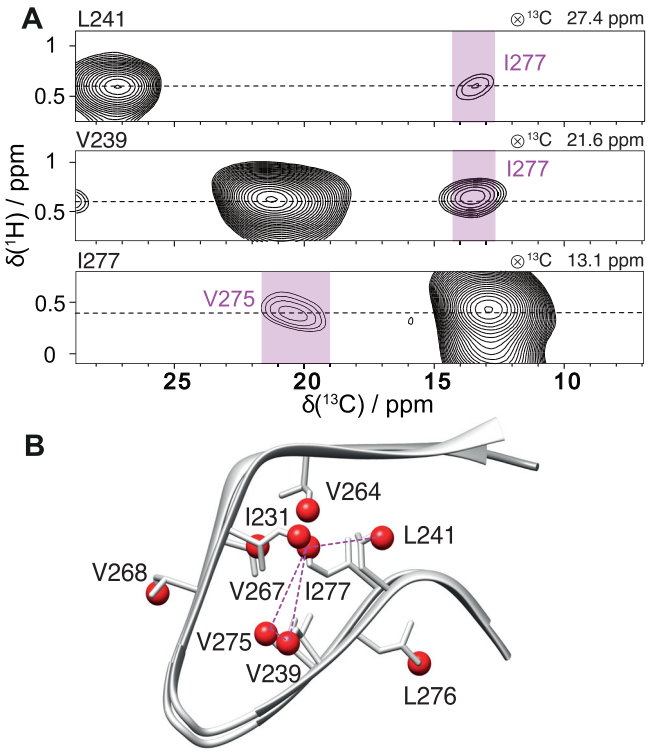


FIGURE 5 | Proton-detected DNP enables structurally informative ^{1}H-^{1}H correlation spectroscopy. (A) Selected two-dimensional ^{1}H-^{13}C planes extracted from proton-detected 3D (^{1}H)-^{13}C-(^{1}H, ^{1}H-RFDR)-^{13}C-^{1}H spectrum, under DNP conditions (experiment time: ~8 h), illustrating representative through-space ^{1}H-^{1}H correlations for the $^{13}CHD_2$-labeled methyl groups of L241 (top), V239 (middle) and I277 (bottom). (B) Corresponding methyl-methyl contacts mapped onto the HET-s(218–289) fibril structure (PDB code: 2KJ3).

to preserve proton resolution under cryogenic DNP conditions. More broadly, they define a new experimental regime in which the sensitivity afforded by hyperpolarization can be constructively combined with the high spectral resolution and intrinsic sensitivity of proton detection. We anticipate that this methodology will substantially extend the reach of biomolecular MAS NMR to challenging systems, including limited-quantity samples, weakly populated states, large supramolecular assemblies, and targets affected by severe spectral crowding.

Author Contributions

James Tolchard: conceptualization, investigation, writing – original draft, writing – review and editing, methodology, validation, software, data curation, visualization, formal analysis. **Tanguy Le Marchand:** conceptualization, investigation, writing – original draft, writing – review and editing, methodology, validation, visualization, formal analysis, data curation. **Nadia El Mammeri:** conceptualization, investigation, writing – original draft, writing – review and editing, validation, methodology, visualization, formal analysis. **Kshama Sharma:** investigation, methodology, validation, writing – review and editing, formal analysis. **Ajit Kumar Bishoyi:** methodology, investigation, formal analysis, writing – review and editing. **Mélanie Berbon:** investigation, methodology, writing – review and editing. **Kristaps Jaudzems:** investigation, methodology, writing – original draft, writing – review and editing. **Dorothea Wisser:** investigation, methodology, writing – review and editing. **Asen Daskalov:** investigation, methodology, writing – review and editing, validation. **Sven J. Saupe:** conceptualization, methodology, validation, writing – review and editing. **Jan Stanek:** investigation, methodology, validation, writing – review and editing. **Zhiyu Sun:** methodology, visualization, writing – review and editing, investigation. **Olivier Ouari:** investigation, methodology, validation, writing – review and editing, resources. **Birgit Habenstein:** investigation, funding acquisition, writing – original draft, writing – review and editing, conceptualization, validation, project administration, supervision, resources. **Anne Lesage:** supervision, resources, project administration, validation, writing – review and editing, writing – original draft, funding acquisition, investigation, conceptualization. **Guido Pintacuda:** conceptualization, investigation, funding acquisition, writing – original draft, validation, writing – review and editing, project administration, supervision, resources. **Antoine Loquet:** supervision, resources, project administration, writing – review and editing, validation, conceptualization, investigation, funding acquisition, writing – original draft.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at https://doi.org/10.5281/zenodo.17803570, reference number https://doi.org/10.5281/zenodo.17803570. *[the link will be active upon publication]*

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

The supporting information contains experimental details, including sample preparation, NMR acquisition parameters, as well as details on the calculation of sensitivity gains. The authors have cited additional references within the Supporting Information.[43–53]

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